

Nextday Wildfire Spread Prediction

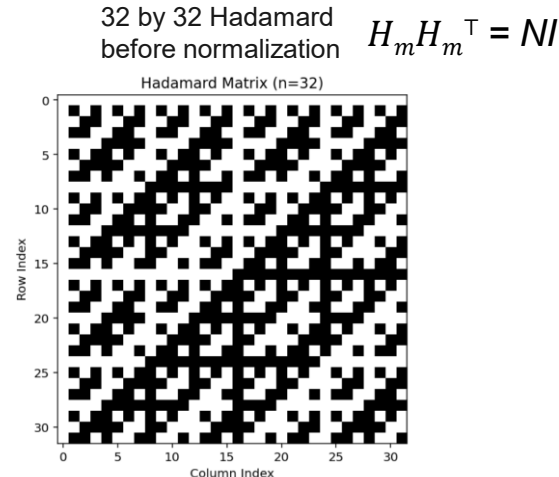
□ Hadamard Transform Matrix

- The Hadamard transform H_m is a $2^m \times 2^m$ matrix composed only of +1 and -1 entries (scaled by a normalization factor).
- It maps the input signal into a latent domain where information is represented as correlations with orthogonal, non-sinusoidal basis patterns
- For 2D inputs, it is applied sequentially along the vertical and horizontal dimensions implemented as $Y = H_m X H_m^\top$

➤ Definition

$$H_m = \frac{1}{2^{m/2}} \begin{pmatrix} H_{m-1} & H_{m-1} \\ H_{m-1} & -H_{m-1} \end{pmatrix}$$

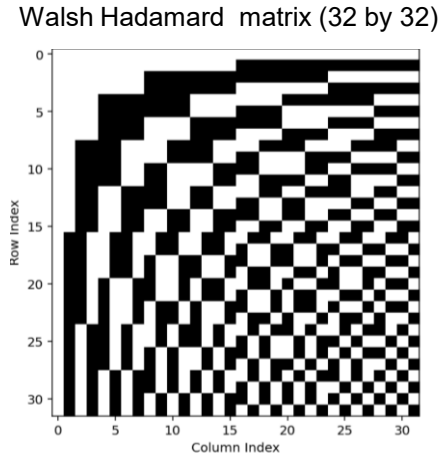
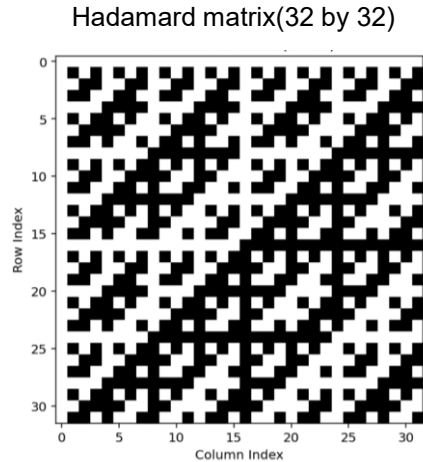
$$H_0 = +(1) \Rightarrow H_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \Rightarrow H_2 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \Rightarrow$$



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❑ Walsh Hadamard Transform Matrix

- Reordering the rows of the Hadamard matrix from low to high frequency.



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❑ Discrete Cosine Transform (DCT) Matrix

- DCT-II is defined by an $N \times N$ orthonormal matrix D whose entries are given by cosine basis functions scaled by appropriate normalization factors.
- For 2D inputs, it is applied sequentially along the vertical and horizontal dimensions as in $Y = CXCT^T$.

➤ Definition

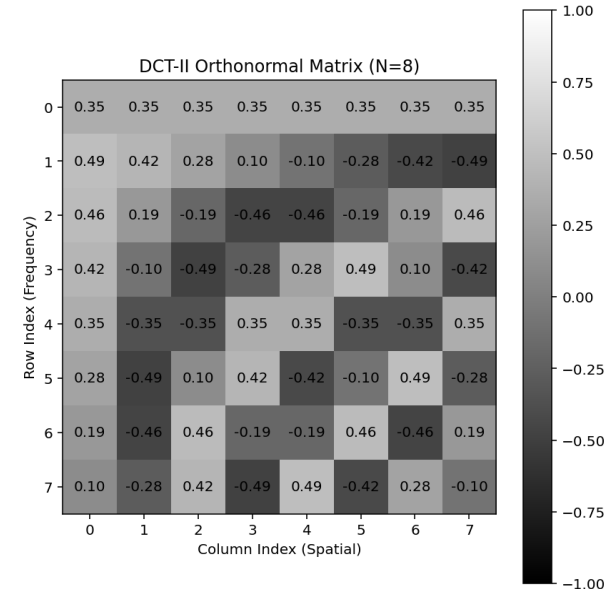
$$C_{k,n} = \alpha_k \cos \left[\frac{\pi}{N} \left(n + \frac{1}{2} \right) k \right]$$

$k \in \{0, 1, 2, \dots, N - 1\}$ ----Frequency Index

Each k corresponds to a cos basis function with increasing frequency

$$\alpha_k = \begin{cases} \sqrt{\frac{1}{N}}, & k = 0 \\ \sqrt{\frac{2}{N}}, & k > 0 \end{cases} \quad \text{----Scaling Factor}$$

$n \in \{0, 1, \dots, N - 1\}$ ----spatial sample index



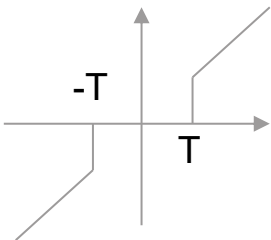
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□ Hard/Soft Thresholding

- The Hard/Soft-thresholding operator with nonnegative **learnable thresholds** T is applied to further compress low-magnitude coefficients.

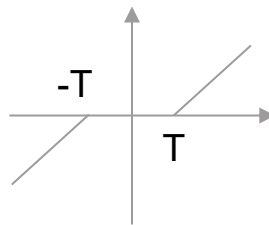
➤ Definition

$$Th_h(e) = \begin{cases} e, & |e| > T \\ 0, & |e| \leq T \end{cases}$$



- Hard thresholding enforces sparsity by setting coefficients with **magnitudes below the threshold to zero**.

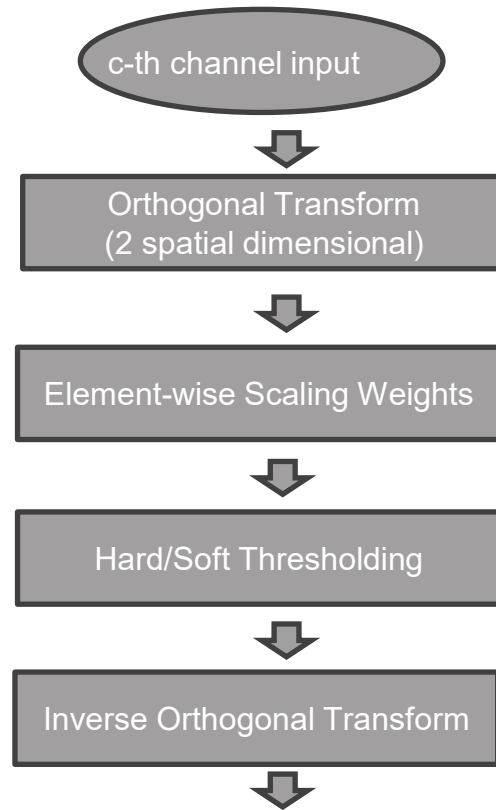
$$Th_s(e) = \begin{cases} \text{sgn}(e) (|e| - T) & |e| > T \\ 0 & |e| \leq T \end{cases}$$



- Soft thresholding enforces sparsity by **performing continuous shrinkage**, smoothly reducing magnitudes toward zero.

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- ❑ Orthogonal perceptron layer



$$\mathbf{X}_c \in \mathbb{R}^{N \times N}$$

$$\begin{aligned} \hat{\mathbf{X}}_c &= \mathbf{H}_N \mathbf{X}_c \mathbf{H}_N & \mathbf{H}_N \mathbf{H}_N^\top &= N \mathbf{I} \\ \text{OR } \hat{\mathbf{X}}_c &= \mathbf{D}_N \mathbf{X}_c \mathbf{D}_N^\top & \mathbf{D}_N \mathbf{D}_N^\top &= \mathbf{I} \end{aligned}$$

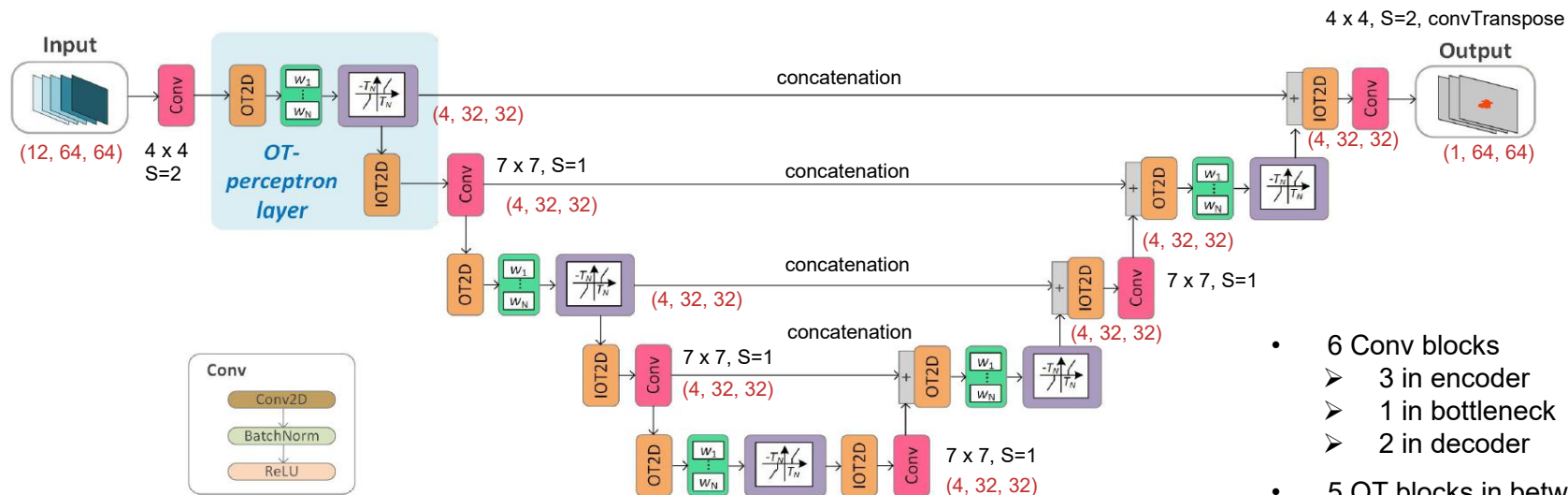
$$E_c(i, j) = W_c(i, j) \hat{X}_c(i, j)$$

$$Th_h(e) = \begin{cases} e, & |e| > T \\ 0, & |e| \leq T \end{cases} \quad Th_s(e) = \begin{cases} \text{sgn}(e) (|e| - T) & |e| > T \\ 0 & |e| \leq T \end{cases}$$

$$\begin{aligned} \mathbf{Y}_c &= \frac{1}{N^2} \mathbf{H}_N \mathbf{Z}_c \mathbf{H}_N \\ \text{OR } \mathbf{Y}_c &= \mathbf{D}_N^\top \mathbf{Z}_c \mathbf{D}_N \end{aligned}$$

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□ Fully integrated Orthogonal Transform-Unet

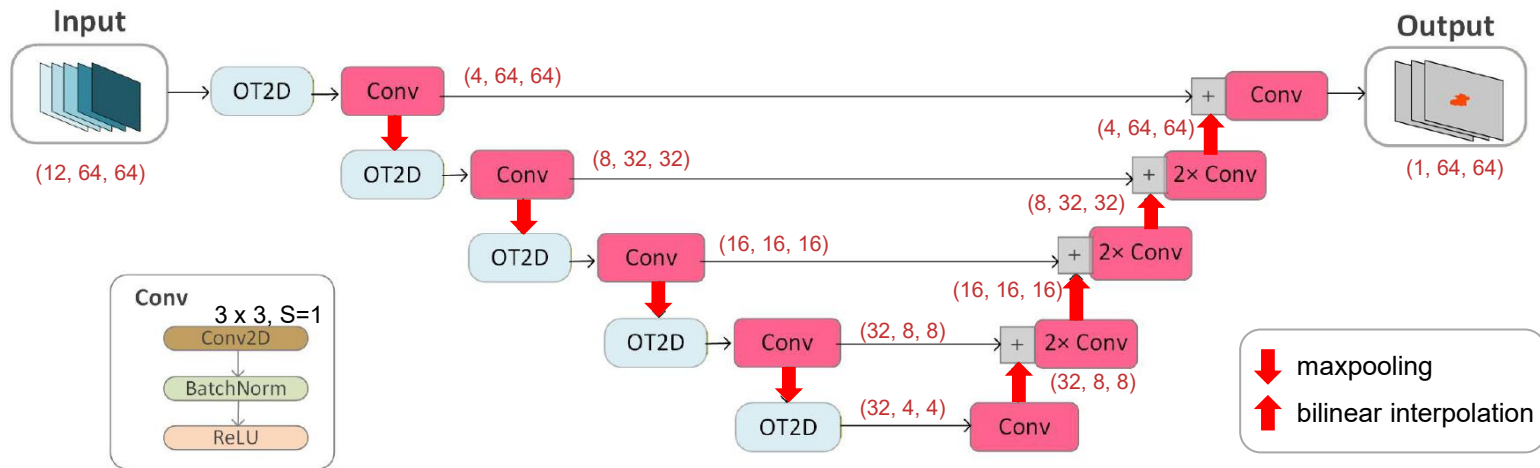


- 6 Conv blocks
 - 3 in encoder
 - 1 in bottleneck
 - 2 in decoder
- 5 OT blocks in between to introduce transform-domain processing

- Passes both the spatial and frequency domains to decoder
- Only spatial change on the boundary

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□ OT-Encoder-Unet

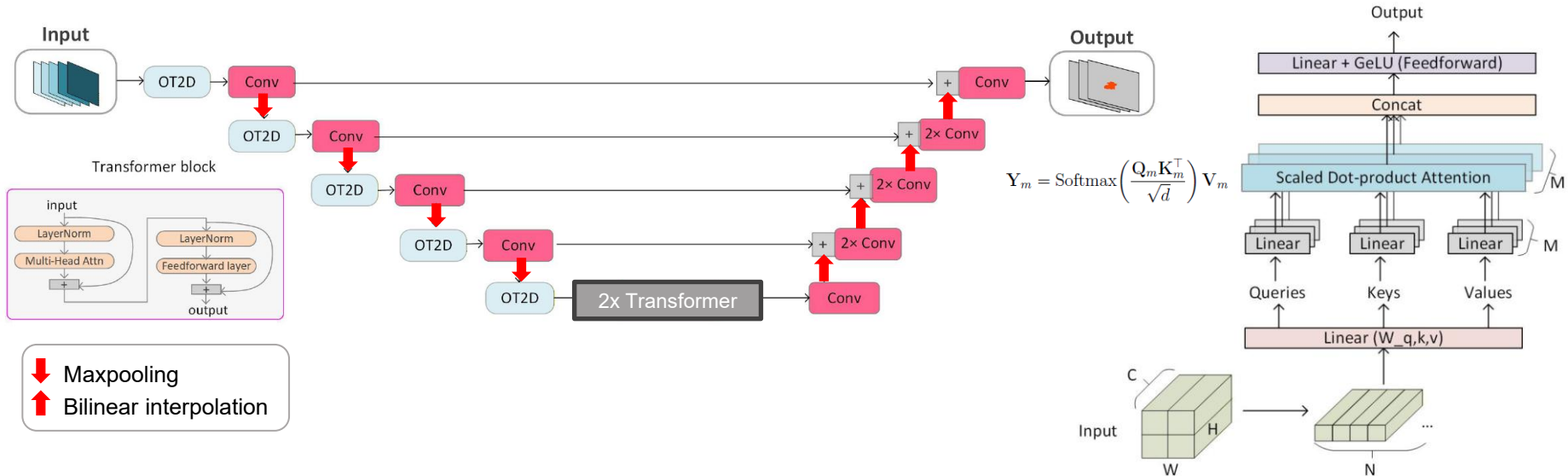


- Replace first conv in each encoder stage of regular Unet with OT2D block
- Continuously lower resolution & increasing channel

- 5 OT2D block
 - 4 in encoder,
 - 1 in bottleneck
- 12 conv layer
 - 4 in encoder
 - 1 in bottleneck
 - 7 in decoder

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OT-Transformer-Unet

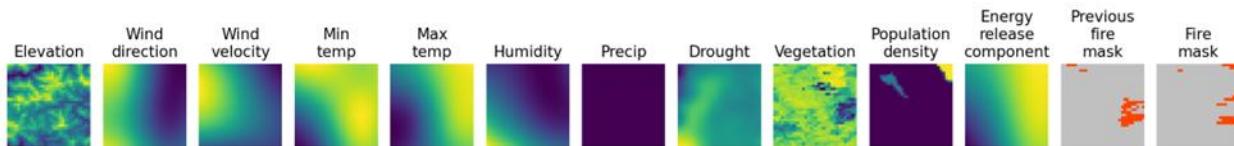


Adds 2 additional transformer blocks into the bottleneck of OT-Enc-Unet

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❑ Experimentation

Dataset: Next-Day Wildfire Spread dataset (Google Research, 2023, baseline CNN autoencoder), predicting both pre-fire and post-fire masks.



Input: 64×64 satellite image with 12 channels.

Output: Probability map; final binary mask obtained using a 0.5 threshold.

Loss Function: Weighted combination of Binary Cross-Entropy (BCE), Dice Loss, Focal Loss.

$$\mathcal{L} = \lambda_{\text{BCE}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{Dice}} \mathcal{L}_{\text{Dice}} + \lambda_{\text{Focal}} \mathcal{L}_{\text{Focal}}$$

Setup: 300 epochs, Adam optimizer, LR 1×10^{-4} .

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❑ Loss Function

$$\mathcal{L} = \lambda_{\text{BCE}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{Dice}} \mathcal{L}_{\text{Dice}} + \lambda_{\text{Focal}} \mathcal{L}_{\text{Focal}}$$

$$\lambda_{\text{BCE}} = 0.4, \lambda_{\text{Dice}} = 0.3, \lambda_{\text{Focal}} = 0.3$$

• Binary cross-entropy (BCE)

- Negative log-likelihood of a Bernoulli random variable

$$y \sim \text{Bernoulli}(p)$$

- Often used for sample-level classification

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N (w y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i))$$

$y_i \in \{0, 1\}$ ----Ground truth label

$\hat{y}_i \in [0, 1]$ ----Predicted possibility

N ----Number of samples (pixels)

w ----Positive weights enhancing positives for class imbalance (set as 3)

• Focal loss

- Builds on BCE

- Adds **confidence-based modifiers** (suppresses easy samples, focuses on hard examples)

$$\mathcal{L}_{\text{Focal}} = -\frac{1}{N} \sum_{i=1}^N (\overset{\text{Positive term modifier}}{\alpha(1 - \hat{y}_i)^\gamma} y_i \log(\hat{y}_i) + (1 - \alpha) \overset{\text{Negative term modifier}}{\hat{y}_i^\gamma} (1 - y_i) \log(1 - \hat{y}_i))$$

$\alpha \in [0, 1]$ ----Fixed for imbalance control

Positive class	Easy positive	$\hat{y} \approx 1$	Loss suppressed	$(1 - \hat{y})^\gamma \approx 0$
	Hard positive	$\hat{y} \ll 1$	Full loss	$(1 - \hat{y})^\gamma \approx 1$
Negative class	Easy negative	$\hat{y} \approx 0$	Loss suppressed	$\hat{y}^\gamma \approx 0$
	Hard negative	$\hat{y} \approx 1$	Full loss	$\hat{y}^\gamma \approx 1$

❖ Assume independent Bernoulli trials (pixel-wise)

❖ Ignore spatial structure

❖ Doesn't optimize intersection over union (IoU)/overlap directly

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- Dice loss

- Derived from Dice Similarity Coefficient (Dice) **directly measures overlap** between predicted and ground-truth in foreground regions:

$$\text{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}|}$$

Y ----ground-truth positives

$\hat{Y}$ ----predicted positives



differentiable formulation

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i y_i \hat{y}_i + \epsilon}{\sum_i y_i + \sum_i \hat{y}_i + \epsilon} \rightarrow \begin{array}{l} 2\text{TP} \\ 2\text{TP} + \text{FN} + \text{FP} \end{array}$$

$y_i \in \{0, 1\}$ ----ground truth pixel

$\hat{y}_i \in [0, 1]$ ----predicted probability

- Does not involve true negatives, therefore robust to **class imbalance** and **background dominance**;
- Often combined with BCE and focal loss to jointly optimize pixel-wise accuracy and region-level overlap

Loss	Key Property	Main Role
BCE	Pixel-wise likelihood	Stable optimization
Focal Loss	BCE with confidence-based weighting	Emphasizes hard pixels
Dice Loss	Region-level overlap; ignores true negatives	Robust to background dominance and class imbalance

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❑ Customized Data Preprocessing

- Random Margin Cropping

- Uncertain pixels (-1) are first set to 0.
- Then we softly perturb both mask classes:

- **Background Pixels (0):**

$$x \sim \mathcal{U}(0.01, 0.03)$$

- **Fire Pixels (1):**

$$x \sim \mathcal{U}(0.8, 0.99)$$

- ✓ Expands data diversity and encourages the model to learn robust, soft boundary representations.

- Gaussian-Mixture Data Smoothing

- Use various gaussian filters with a set of predefined standard deviation

$$\mathbf{G}_{\sigma_k} = \text{GaussBlur}(\mathbf{M}, \sigma_k)$$

{0.4, 0.8}

- Mean aggregation:

$$\mathbf{P} = \frac{1}{K} \sum_{k=1}^K \mathbf{G}_{\sigma_k}$$

- ✓ Improve spatial coherence and handle missing information in the mask (M)

Applied on sparse maps with missing data and highly imbalanced binary masks (wind maps and pre-fire masks)

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□ Results

Model	Params	Prec. (%)	Recall (%)	IoU (%)	F1 (%)	$\Delta F1_1$ (pp)	$\Delta F1_2$ (pp)
AECNN (Baseline in paper)	77,264	66.08	62.63	40.95	64.31	–	–
LiteUnet (Designed baseline)	53,205	86.18	59.03	51.67	70.07	–	–
HT-Unet	44,989	82.49	58.49	46.20	68.45	+4.14	-1.62
HT-Unet (Trainable HT)	47,037	76.39	59.93	46.01	67.16	+2.85	-2.91
HT-Enc-Unet	42,477	89.67	57.86	53.32	70.34	+6.03	+0.27
DCT-Enc-Unet (No compression)	42,477	87.48	58.64	50.81	70.21	+5.90	+0.14
DCT-Enc-Unet (50% compression)	34,293	77.72	61.52	50.70	68.68	+4.37	-1.39
HT-Trans-Unet	92,269	86.54	58.69	52.99	69.94	+5.63	-0.13
DCT-Trans-Unet	92,269	82.20	60.62	50.03	69.78	+5.47	-0.29

Param reduc. 41.77%, 15.44%

Param reduc. 45.02%, 20.17% 

Param reduc. 55.61%, 35.55%

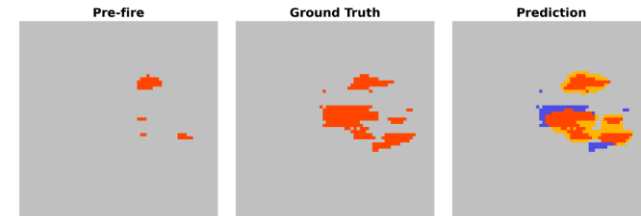
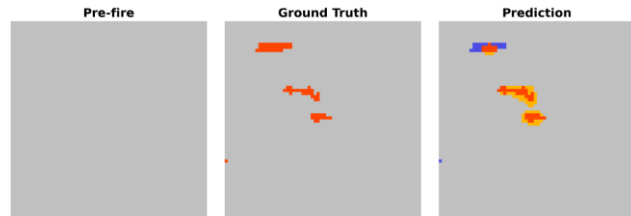
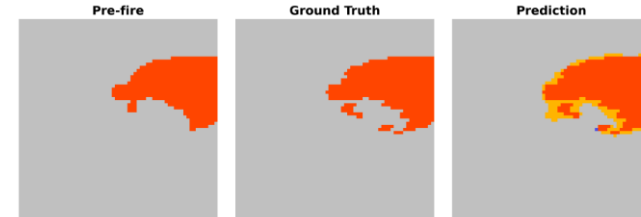
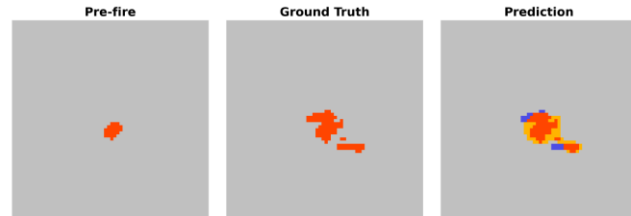
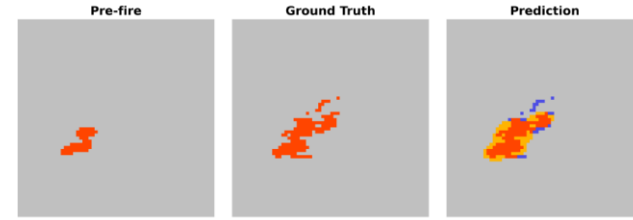
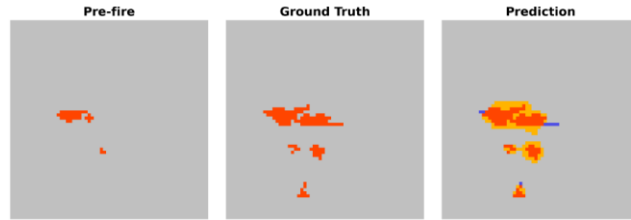
All evaluation metrics are reported in percentages. $\Delta F1_1$ and $\Delta F1_2$ denote absolute differences in percentage points (pp) relative to the AECNN baseline ($F1 = 64.31\%$) and the Regular Light Unet baseline ($F1 = 70.07\%$), respectively.

❏ Conclusion

- Key Findings (OT-UNet Family)
 - **Parameter efficiency:** OT-UNet models significantly reduce parameters while improving F1 overall.
 - **Precision gains:** All variants show consistent precision improvements, indicating fewer false positives.
 - **Recall behavior:** Recall gains not shown, expected due to severe class imbalance in the Next Day Wildfire Spread dataset.
 - **Best model size & accuracy trade-off (OT-Encoder-UNet):**
 - HT-Enc-UNet: largest F1 gain with fewer parameters than AECNN and LiteUNet.
 - DCT variants: Enable further parameter reduction via spectral compression while maintaining comparable F1.
 - **Transformer Bottleneck:**
 - Larger parameter increase with lower performance gain.
 - Attention is more effective in larger models and datasets.
 - In lightweight models, spectral filtering already captures global structure, limiting attention benefits.
 - Future work: more efficient attention integration (e.g., hierarchical tokens, tighter spectral–attention coupling)

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❑ Prediction examples



Yellow: false positives (overpredicted areas); Blue: false negatives (missed fire regions)

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□ Additional Evaluation on WildfireSpreadTS dataset

Input: 128×128 with 40 channels

Output: Day $t+1$ wildfire mask

Loss: Dice loss is used to be consistent with original paper

Setup: 200 epochs, Adam optimizer, LR 1×10^{-4}

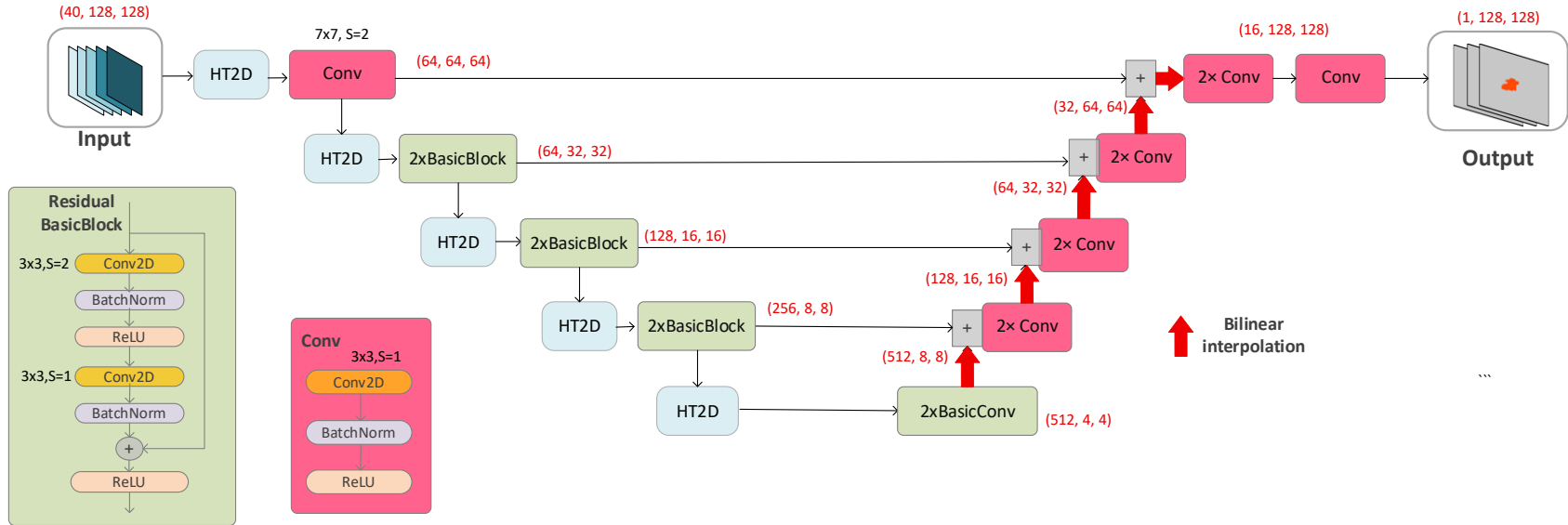
Future work will focus on extending proposed architecture to model multi-day time-series data, as provided in the WildfireSpreadTS dataset.

Model		Params	Precision (%)	Recall (%)	IoU (%)	F1 (%)
Baseline	UNet	14.36M	53.55	65.40	41.73	58.89
(ResNet18	back-					
bone)						
HT-Resnet18-UNet		14.43M	57.04	61.18	41.88	59.04
(Resnet18	Back-					
bone)						
HT-UNet		0.17M	51.02	64.64	39.89	57.03
HT-Enc-UNet		0.12M	51.41	66.69	40.90	58.06

Over two orders of magnitude fewer parameters with comparable accuracy
Adding HT2D perceptron layer as a header for each stage of Resnet encoder improves F1

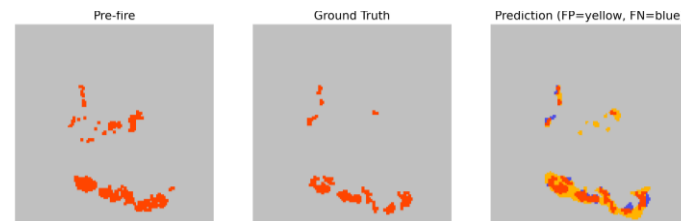
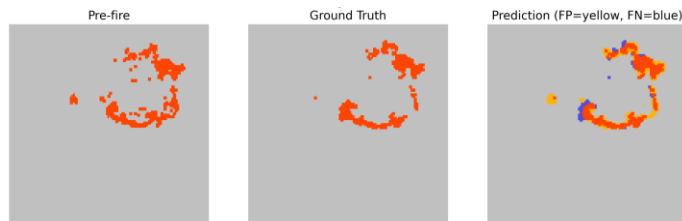
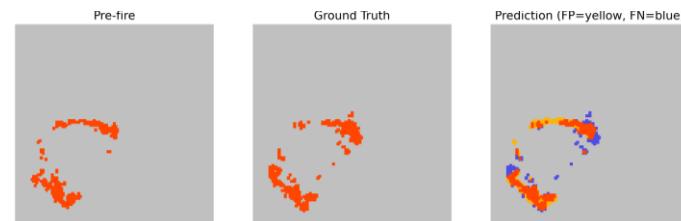
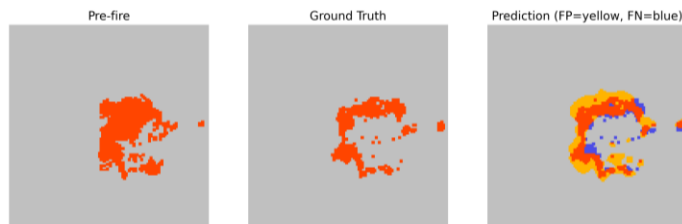
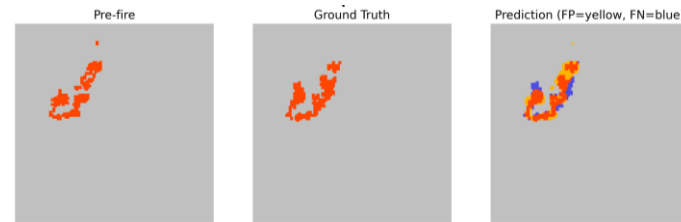
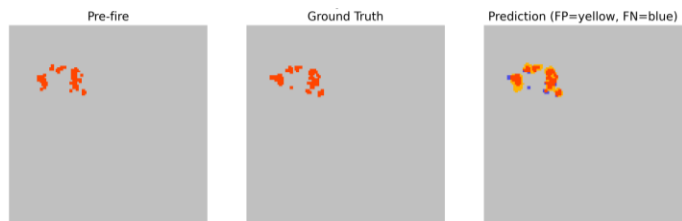
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HT2D-Resnet18-Unet



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❑ Prediction examples on WildfireSpreadTS dataset



Yellow: false positives (Overpredicted areas); Blue: false negatives (missed fire regions)